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EFFECT OF PRACTICE DISTRIBUTION AND EXPERIENCE ON THE PERFORMANCE AND RETENTION OF A DISCRETE SPORT SKILL[1]

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Summary.—This study examined how practice distribution influenced performance and learning of a discrete sport skill, the Australian Football (AF) handball pass. A secondary aim was to assess whether previous experience playing competitive Australian Football influenced learning. Participants performed the handball 50 times (5 blocks x 10 repetitions) using either a massed (1 sec. between repetitions or distributed (30 sec. between repetitions) practice schedule. Testing consisted of pre-test, acquisition, immediate retention (10 min.), and delayed retention (2 weeks) sessions. Performance accuracy scores improved in the massed practice condition from pre-test to immediate retention and from pre-test to delayed retention. Likewise, performance improved in the distributed practice group from pre-test to immediate retention, but scores were not different from pre-test to delayed retention, and decreased from immediate retention to delayed retention. While students with previous AF experience performed better overall, there were no differences between the massed and distributed groups based on experience. Results suggested that, regardless of previous related skill, massed practice of a discrete sport skill may lead to better retention of learning over a two-week period.

To maximise motor skill learning and performance, it is important for researchers and practitioners to effectively manipulate variables within the practice environment. Massed and distributed practice schedules are commonly referred to when comparing the ratio of rest to work done during practice. Although the terms are relative, when applied to the inter-trial interval, massed practice generally refers to a practice schedule where the amount of rest between trials is very short and distributed practice refers to a schedule where the amount of practice between trials is relatively long (Magill, 2011). The influence practice distribution has on skill performance

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and learning has long been a topic of investigation in motor learning, yet remains an area of controversy (Magill, 2011). The majority of research on practice distribution effects has been conducted on continuous skills, with only a few studies examining discrete, lab-based tasks; as such, we understand little about how practice distribution influences learning and performance of discrete skills (Schmidt & Lee, 2011). The use of lab-based activities, as opposed to everyday tasks or sport-specific skills, also limits our ability to generalize these findings to more common movement-learning situations such as fundamental movement skills (e.g., kicking, throwing), sports skills, or tasks where learners may have prior experience.

The common finding from the distribution of practice literature is that distributing practice results in better performance. In their review of 47 papers (52 effect sizes), Lee and Genovese (1988) reported that distributed practice schedules resulted in greater performance improvements during acquisition (d = 0.96), and better retention (d = 0.53) when compared to massed practice schedules. The variety of ways retention and transfer are defined in the studies reviewed called into question the strength of Lee and Genovese's conclusions regarding retention (Christina & Shea, 1988; Newell, Antoniou, & Carlton, 1988; Donovan & Radosevich, 1999). For example, Lintern (1988) suggested the distinction between immediate and delayed retention is essential, as tests that are delayed by less than 20 minutes are not likely to distinguish performance effects from learning effects. A subsequent meta-analysis conducted by Donovan and Radosevich (1999) reviewed 63 studies (112 effect sizes) and found a similar benefit for using distributed practice schedules over massed schedules (d = 0.46). Separate effect sizes were also calculated for immediate retention (i.e., task performance immediately after practice; d = 0.45), and delayed retention performance, (i.e., task performance that was separated from practice a minimum of 24 hours; d = 0.51). Given the limited number of studies that used a delayed retention test—effect sizes from 16 studies contributed to the overall effect size for retention, and of these studies, 13 used a retention interval of 24 hours and just 3 were greater than 24 hours—the long-term effects of massed vs distributed practice should be further investigated to compare temporary performance effects with more permanent learning effects.

The type of task may also influence the benefit observed when distributing practice; continuous motor tasks have arbitrary beginning and ending points (e.g., rotary pursuit tasks), whereas discrete tasks have a clearly defined beginning and end (e.g., throwing a ball). Many fundamental movement and sporting tasks (e.g., kicking, throwing, striking, and catching) are discrete skills. Lee and Genovese (1988) reported that the literature had focused almost exclusively on continuous motor tasks, with only one study in their meta-analysis using a discrete skill. Carron (1969) investigated the learning of a peg-turning task and suggested similar or slightly better learning for massed practice than distributed practice. Lee and Genovese (1989) later replicated and extended Carron's study by using a discrete task of even shorter duration and compared acquisition and retention with a continuous skill. Consistent with previous studies, they reported that distributed practice was better for acquisition and retention for the continuous task; however, for the discrete task, massed practice was better. The benefit of using a massed practice schedule when performing a discrete skill is that there is less fatigue so the learner can practice a greater number of repetitions in a shorter amount of time. In applied settings (e.g., physical education classes, rehabilitation), where time and resources are often limited, massed practice may have the added benefit of allowing more effective use of time.

A lack of theoretical development explaining the benefits of distributed practice for most activities appears to have slowed research progress (Schmidt & Lee, 2011). Explanations put forward include fatigue, cognitive effort, and memory consolidation (Magill, 2011). The fatigue explanation is that massed practice schedules, because of their short rest intervals, can induce fatigue, which could negatively influence learning. Massing practice may also reduce the amount of cognitive effort the learner uses on each practice trial, because the task becomes very repetitive. Memory consolidation suggests that to store information effectively in memory, some rest is necessary for biochemical changes to occur. Providing additional rest may assist this to happen; however, this explanation appears to be more applicable to the distribution of practice sessions across days, than the spacing of trials within a session, since the biochemical changes are unlikely to occur in the time between trials, even if trials are distributed.

Due to the lack of research in the area, Schmidt and Lee (2011) concluded that the effect of distributing practice trials for discrete skills was not well understood and recommended that further studies should focus on exploring the effects of massed and distributed practice conditions for discrete skills that are common to sports activities (e.g., kicking, throwing). A few studies have explored discrete sport skills, but have focused on the distribution of practice sessions rather than the inter-trial interval. For example, Singer (1965) compared the effects of distributing practice when learning a novel ball-bouncing skill and found that at the end of the acquisition period, the group that had distributed practice across days performed better than the group who performed all trials in a single session; however, after one month (i.e., retention), the massed practice group's performance was superior to that of the distributed practice group. Dail and Christina (2004) compared the effects of practice distribution for learning a golf-putting task while measuring retention at 1, 7, and 28 days. The massed group practiced all putts in a single session, while the distributed group practiced 60 putts in each session across 4 consecutive days. Consistent with previous findings on continuous skills, distributed practice resulted in better acquisition and retention compared to massed practice. While these studies demonstrate the influence of the practice distribution across sessions, they provide limited understanding of the effects of manipulating the inter-trial interval on the learning of a discrete sport-specific skill.

Task complexity and the learner's experience may also influence the effect of practice distribution. The majority of research in the area has used novel, lab-based motor skills (e.g., tracking, inverted alphabet printing, and mirror tracing tasks) as opposed to applied, sport-specific skills (Lee and Genovese, 1988). In their meta-analyses, Donovan and Radosevich (1999) included measures of overall complexity as well as the mental and physical requirements of the task. Distributed practice had the strongest effect ($d = 0.96$) on simple tasks, while tasks high in overall complexity and physical requirements, but low in mental requirements (e.g., gymnastics), or high in mental requirements (e.g., airplane control simulation) had small effect sizes ($d = 0.11$ and 0.07, respectively). This suggests the type of task may mitigate the practice distribution effect, and raise questions about whether the effects found for simple, lab-based tasks apply to more sport specific movements and skills. The emphasis on novel, lab-based activities in the literature has also made it difficult to assess whether practice-distribution effects apply to both inexperienced and experienced learners. Experience often influences the extent that other practice effects, such as contextual interference (Ollis, Button, & Fairweather, 2005), have on learning, which suggests that it may influence the effectiveness of distributing practice trials; yet, to our knowledge, no previous studies have examined this effect.

The distribution of practice trials is an important practice variable in the learning of motor skills. Research on continuous skills shows that distributed practice leads to better performance (acquisition) and learning (retention) compared to massed practice. While the limited research on discrete skills has indicated that massed practice may be more effective than distributed practice, the reliance on lab-based activities, as opposed to common activities or sport-specific skills, has limited knowledge of the effects of practice distribution in more common movement learning situations. Thus, we know little about applying principles of practice distribution to common discrete movement skills (e.g., kicking, throwing) and what influence prior experience has on practice distribution effects. This study compared the effects of prior experience and also massed and distributed inter-trial practice intervals on the learning of a discrete sport skill. By examining a discrete sport-specific skill, this study should provide insight into the application of practice schedules in designing practice sessions and training programs for skill acquisition. Consistent with previous research:

Hypothesis. Massed practice schedule would be more beneficial for performance (immediate retention) and learning (delayed retention) of a discrete sport skill. It was also expected that these effects would be more pronounced in a group of less experienced participants.

Method

Participants

Participants were 175 university undergraduate students (99 men, 76 women) ranging in age from 18 to 51 years ($M = 20.7$, $SD = 3.3$). Most participants reported that they had played Australian Football (AF) ($n = 128$), with the majority playing at a competitive level (e.g., for a club or school) ($n = 112$). Using quasi-random selection (i.e., participants formed groups of 3–4 and each group was randomly assigned to a condition) participants were assigned to one of two practice conditions: massed practice ($n = 92$; 46 men, 46 women) and distributed practice ($n = 83$; 53 men, 30 women). Participants were classified as experienced or inexperienced (*post hoc*) based on whether they had reported playing AF competitively (e.g., for school or a club) or not. According to this classification, 112 participants were categorised as experienced and 63 participants were classified as inexperienced. There were 53 experienced participants and 28 inexperienced participants in the massed condition and 59 experienced participants and 35 inexperienced participants in the distributed condition. The study was approved by a University Human Research Ethics Committee, and all participants signed an informed consent form before beginning the study.

Task

The discrete sport skill selected for this experiment was an AF handball pass, which required participants to do a handball pass of a regulation Australian Football League (AFL; 2007) football at a target fixed on a wall. The handball pass is a technique that involves holding the ball in one hand while striking it with a clenched fist created by the other hand (Austrailan Football League, 2007). The target consisted of 5 concentric circles, and participants were awarded 5 points for hitting the centre of the target, decreasing 1 point per circle as they hit further from the centre. No points were awarded for missing the target area. The outermost target circle had a diameter of 80 cm, decreasing to 64 cm, 48 cm, 32 cm, and 16 cm for the central target. Each pass was from a distance of 5 m and was completed indoors in a gymnasium with a flat wooden floor. A ball was handed to the participant prior to each pass.

Pilot testing was conducted to choose the passing distance and target size to manipulate the difficulty of the task. The aim in pre-setting the difficulty was for beginner performers to achieve a score of approximately 50–60% of maximum in initial trials, thus creating a sufficiently difficult task that there would be adequate opportunity for improvement due to practice.

Procedure

Each participant completed two sessions, which were conducted two weeks apart. In the first session, the researchers explained the study protocol and students were assigned to an experimental condition. Participants in the massed practice condition completed 50 practice trials with an inter-trial interval of 1 sec. Participants in the distributed practice condition completed 50 practice trials with an inter-trial interval of 30 sec. Participants first completed a pre-test of 10 trials with a 10-sec. inter-trial interval and then practiced the handball pass in either the massed or distributed practice condition. Participants completed five practice blocks of 10 handball passes to the target with two minutes between blocks (50 trials total). They then completed an immediate retention test of 10 trials with a 10-sec. inter-trial interval after 10 minutes. In the second session, conducted two weeks later, participants completed a delayed retention test of 10 trials (10-sec. inter-trial interval). All inter-trial intervals were timed using a handheld stopwatch. The inter-trial interval was the time from when the ball hit the target until the participant was instructed by the researcher to perform the next pass with the instruction "go." A score for performance was provided to participants as verbal feedback on completion of each trial during practice and testing.

Data Analysis

Accuracy scores during testing were averaged by block. A 2 (practice condition) X 2 (experience) X 3 (time) mixed-model analysis of variance (ANOVA) was calculated to evaluate the effects of practice condition (between-groups) and experience and on performance accuracy scores at pre-test, immediate retention test, and delayed retention test (within-groups). Pairwise comparisons using the Bonferroni correction were conducted to follow up statistically significant effects. While gender was not a dependent variable of interest in the context of this experiment it is worth noting that, as a group, men out-performed women on accuracy scores although there were no effects (practice condition, experience, or time) beyond this.

Results

Table 1 presents the performance results for the experienced and inexperienced groups at each session. There was a statistically significant difference between scores across sessions ($F_{2, 342} = 26.90$, $p < .001$, $\eta_p^2 = 0.14$). Scores improved between pre-test and immediate retention ($p < .001$), between

TABLE 1

MEAN SCORES FOR EXPERIENCED AND INEXPERIENCED PARTICIPANTS IN MASSED AND DISTRIBUTED PRACTICE CONDITIONS AT PRE-TEST, POST-TEST, AND RETENTION TEST

Condition	Pre-test		Immediate Retention		Delayed Retention	
	M	*SD*	*M*	*SD*	*M*	*SD*
Experienced-massed	2.50	0.86	2.80	0.81	2.64	0.84
Experienced-distributed	2.50	0.71	3.05	0.75	2.63	0.66
Inexperienced-massed	1.41	0.91	1.81	0.94	1.89	0.98
Inexperienced-distributed	1.65	0.80	2.25	0.95	1.83	0.89

pre-test and delayed retention ($p < .001$), and decreased between immediate and delayed retention ($p < .001$). Experienced participants had statistically significantly higher scores on the passing task than inexperienced participants ($F_{1, 171} = 65.75$, $p < .001$, $\eta_p^2 = 0.28$). There was no significant experience X time interaction ($F_{2, 342} = 1.32$, $p = .27$, $\eta_p^2 = 0.01$), thus the pattern of change for the experienced and inexperienced participants did not differ across time.

Performance scores for the massed and distributed practice conditions are presented in Fig. 1. There was a statistically significant practice condition X time interaction ($F_{2, 342} = 4.76$, $p = .009$, $\eta_p^2 = 0.03$), indicating that the pattern of change for the two practice conditions was different across sessions. The massed practice condition improved from pre-test to immediate retention ($p < .001$) and from pre-test to delayed retention ($p = .003$). They also maintained performance levels, as there was no significant decline in performance between immediate and delayed retention scores ($p > .90$). The distributed group's performance improved from pre-test to immediate retention ($p < .001$), but scores were not statistically significantly different from pre-test to delayed retention ($p = .16$), and decreased from immediate to delayed retention ($p < .001$). Comparisons between groups at each session revealed no differences between groups at the pre-test and delayed retention test, however the distributed group performed statistically significantly better ($p = .01$) on the immediate retention test. There was no statistically significant practice condition X experience X time interaction ($F_{2, 342} = .75$, $p = .47$, $\eta_p^2 = 0.004$), indicating that, across testing sessions in both practice groups, the accuracy scores followed a similar pattern of change for both experienced and inexperienced participants.

DISCUSSION

This study compared the effect of practice distribution and experience on learning a discrete sport skill. Literature on the inter-trial interval has generally found that longer periods of rest between trials are beneficial for

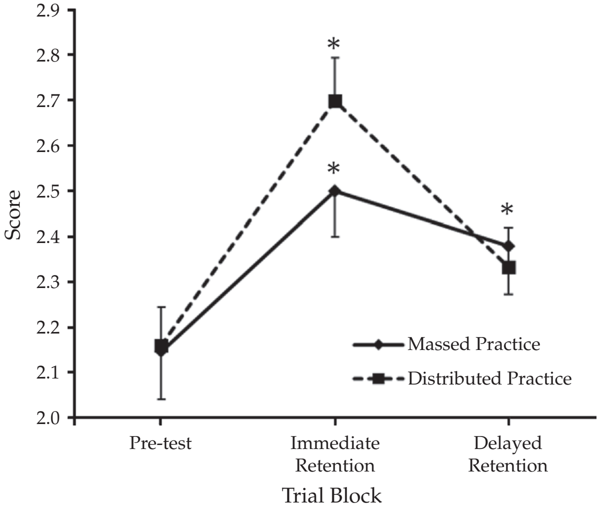

FIG. 1. Mean score for massed and distributed practice conditions (collapsed across experience levels) at pre-test, immediate retention (10 min.), and delayed retention (2 wk). *Performance significantly different from pre-test, $p < .05$.

the learning of continuous sport skills (Lee & Genovese, 1988; Donovan & Radosevich, 1999). Understanding of the effects of practice distribution on discrete sport skills, as well as the influence of prior experience, is limited. Research is needed to provide recommendations for practitioners. Results from this study suggest that massed practice led to improved skill learning (relative to initial performance levels) and prior experience did not influence the distribution effect. Massed practice scores improved from pre-test to immediate and delayed retention, and there were no differences between the immediate and delayed retention tests. Distributed practice scores showed a similar increase from pre-test to immediate retention; however, there was a decrease in scores from immediate to delayed retention and no difference between scores on the pre-test and delayed retention test. Differences between groups on the delayed retention test were not statistically significant, however. So while these findings provide some initial evidence that, regardless of skill level, massed practice may be useful for promoting long-term retention (up to 2 weeks) of discrete, sport-specific skills, additional research is needed to provide a clearer understanding of these effects.

While the benefits of using massed practice for discrete skills have been demonstrated before (Lee & Genovese, 1989) this is the first study, to our knowledge, that demonstrated some potential benefits for using massed practice in the long-term retention of a sport-specific skill. Although the two meta-analyses conducted in the area of practice distribution almost exclusively reported findings for continuous skills, rather than discrete skills (Lee & Genovese, 1988; Donovan & Radosevich, 1999), both emphasised the importance of considering the performance versus learning effects and highlighted the need for including retention tests. In this study, scores improved in a similar fashion between the groups through immediate retention and, although the absolute difference in scores was not statistically significant at delayed retention, the massed practice group performed slightly better and still maintained accuracy scores that were significantly higher than the pre-test after a 2-week delay. Massed practice is often viewed negatively because it is thought to lead to boredom or fatigue, but from a practical standpoint, using massed practice may be beneficial for practitioners with limited time to teach skills. Another issue that arises is that measuring performance immediately after distributed practice could cause an overestimation of the learning effect, with differences in learning effects between conditions being short-term (Adams & Reynolds, 1954). The use of a retention test of a substantial time period in the current study (2 weeks) and the change in performance scores from post-test to retention test highlight the need for studies exploring distribution of practice effects to compare immediate performance effects (post-test) with more long-term learning effects (retention).

The extensive use of lab-based tasks to explore learning effects (Lee & Genovese, 1988; Donovan & Radosevich, 1999) also means learners are not generally familiar with the skill. Therefore, little is known about the effects of distributing or massing practice trials in relation to experience on the task. In an applied setting, such as rehabilitation or physical education, however, skills are rarely completely new and have often been seen or practiced before, even extensively. In the current study, experienced participants performed better on the sport skill than inexperienced participants, as would be expected. Surprisingly, the pattern of change for the experienced and inexperienced participants was not different across time. In addition, the effect of massed and distributed practice did not differ between the experience levels, indicating that amount of learning was similar for experienced and inexperienced participants. Therefore, it would seem that the recommendations for massed and distributed practice for a discrete skill are similar regardless of an individual's experience on the skill. Further studies with different experience levels are warranted to determine whether the experience findings are consistent.

One interesting feature of the results was the difference in performance scores between the massed and distributed groups at the immediate retention test. Even after performing a limited number of trials, the distributed group performed better, which could indicate there are immediate benefits associated with distributed practice. A better understanding of this effect would have an important benefit in application—the results provide initial evidence that massed practice may be useful for learning discrete skills, but the improvement observed in the distributed practice group on immediate retention suggests that distributed practice of discrete skills may have some benefit. For example, when warming up or preparing for a competitive sporting event, distributed practice may be more beneficial to improve performance on that day, whereas massed practice may be a more appropriate schedule at training sessions (Spittle, 2013). Another consideration is the amount of time available to practice a skill. Because of the longer intertrial intervals under a distributed practice schedule, it takes longer to complete the same number of trials as it does under a massed practice schedule (Schmidt & Lee, 2011). This can reduce the amount of practice engaged in during a practice session (Edwards, 2011), although fatigue and motivation must be taken in account. While speculative at this point, these suggestions do warrant further exploration.

There are potential limitations of the current study that should be considered when interpreting the findings and conclusions. Although the findings are encouraging, the number of repetitions performed in the sessions was small and future research to examine the optimum number of trials to elicit practice distribution effects is warranted. In a game of AF, players handball pass to teammates while under pressure and on the move from different distances, and the current study required participants to handball pass while stationary to an unmoving target at a specified distance, which may limit generalizability. The task did appear to be appropriate, because there was improvement in performance with practice, and experienced participants did perform better than inexperienced participants. Further research using more sport-specific skills may assist in assessing the transfer of learning into more realistic competitive performance (Lee & Genovese, 1988; Donovan & Radosevich, 1999). A relatively broad categorization was used and distinctions between higher and lower level competitors may have been missed; however, both groups responded to the intervention in a similar manner. While we did not find a difference between skill levels, further work is required to support or refute this contention.

A better understanding of the interaction between experience and practice distribution would help coaches and teachers create optimal practice schedules with athletes and students from different competitive skill levels. For example, in the practice variability literature, contextual interference is generally seen to be beneficial for learning. Proposals such as the challenge-point framework (Guadagnoli & Lee, 2004), however, contend that for performers with lower skill, reducing the contextual interference may benefit learning.

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